

Beyond the Grip: AlphaHand for Individual Finger Movement Identification from Consumer EEG

Jonathan Davanzo

First author and lead researcher
Summit High School
jonathandavanzo@alphahand.org

Liam Bales

Summit High School
liam.c.bales@gmail.com

Oliver Buchanan

Summit High School
oliverpbuchanan@icloud.com

Abstract—Consumer-grade EEG headsets offer an accessible path to motor-assistive brain-computer interfaces, but their low channel count, high noise floor, and indirect coupling to hand kinematics make finger-level decoding an open challenge. For individuals who cannot use EMG-based systems due to muscle damage, nerve injury, or other neuromuscular conditions, no consumer EEG alternative has demonstrated deployment-gated finger-level accuracy and replay evidence for practical assistive-control development. AlphaHand is a low-cost Muse 2 pipeline that achieves, for the first time in consumer EEG, subject-specific finger-level command decoding relevant to fine-motor assistive access. The pipeline records timestamp-aligned EEG with event markers, extracts 0.250 s windows stepped at 0.05 s, and trains a calibrated multi-head CNN+LSTM classifier evaluated through the same command gates used for downstream robotic-hand actuation. The classification task spans 11 classes—five fingers (thumb through pinky) each labeled across two movement states (open and close), plus a shared REST class. The featured subject-specific deployment checkpoint (2-M16), trained on two distinct movement sessions and one auxiliary REST session, achieved 89.79% held-out action accuracy, 85.96% deployment-consistent finger accuracy, and 84.66% joint action–finger accuracy under deterministic evaluation. In pseudo-live replay across 3324 emitted positive windows, the command path reached 95.37% non-REST precision and detected 91.11% of 596 movement events at 83 ms median first-command latency; window-level send recall of 31.56% reflects stability gating and command-policy behavior, with classifier error captured separately in the held-out window metrics. False actuation during REST was rare, occurring on only 6 of 2404 (0.25%) true REST windows. Prospective multi-subject testing is the next validation step. Note: this testing will happen before full publishing.

I. INTRODUCTION

Millions of people in the United States live with limb loss or congenital limb difference. Recent analyses estimate the affected population exceeds 5.6 million, including nearly 2.3 million with limb loss and 3.4 million with congenital difference [1]. Limb amputation carries substantial psychological burden alongside its physical consequences; a 2024 meta-analysis estimated pooled depression prevalence after amputation at 33.85% (95% CI: 27.15–40.54%) [2]. Restoring meaningful hand function for this population requires more than binary grasp detection: many daily tasks depend on which finger moves and when, making finger-level intent decoding a demanding and clinically meaningful target.

EMG is the standard control modality for practical upper-limb prostheses [3]. It is effective when informative residual musculature is available. For users with limited or absent

peripheral control signals, consumer EEG can make neural control viable where residual musculature cannot support EMG, opening an access path independent of peripheral motor signals. Low-channel EEG is spatially blurred and sensitive to motion and impedance variation. Achieving finger-level decoding under these constraints is therefore more demanding, and the results are correspondingly more significant. AlphaHand demonstrates that a low-cost, low-channel system can preserve enough usable intent structure to support a conservative assistive-control command path. Finger-level decoding from such hardware therefore represents a stringent demonstration that low-cost brain-computer interfaces [4] can move beyond coarse command vocabularies.

Prior EEG work has shown that classification performance improves substantially with deep architectures, spanning approaches from shallow features to end-to-end learned representations [5], including the Deep ConvNet family and the compact EEGNet architecture [6], [7]. At the task level, noninvasive studies have demonstrated that individual finger movements can be discriminated from EEG under controlled conditions [8], [9], and Xiao and Ding further showed that spectral-structure features from high-density EEG carry discriminative information for individual fingers beyond classic mu/beta features [10]. More recently, Ding et al. demonstrated real-time noninvasive robotic hand control at individual finger level using 128-channel EEG in experienced BCI users [11]. These results establish that EEG contains task-relevant finger information. AlphaHand extends that evidence into the constrained deployment setting: high-density sensing is replaced by a four-channel consumer headset validated for research use [12], [13], and predictions remain reportable after calibration, safety gates, and replay through the command path. Because transfer across users remains a persistent challenge in EEG-BCI, subject-specific deployment is the appropriate first validation target [14]. Additionally, once a classifier is connected to hardware, calibration and uncertainty become major operational concerns [15], [16].

AlphaHand answers the central deployment question by showing that subject-specific structure survives calibration, safety gating, and replay through an auditable command path. This paper makes four contributions:

- An end-to-end consumer EEG pipeline for finger-level command decoding, spanning Muse 2 acquisition, timestamp-aligned labeling, fixed-window extrac-

TABLE I
CONTEXT AGAINST PRIOR NONINVASIVE EEG FINGER-MOVEMENT STUDIES. METRICS REFLECT DIFFERENT TASK DEFINITIONS, SENSORS, AND VALIDATION PROTOCOLS AND SERVE AS CONTEXT ACROSS NON-IDENTICAL EVALUATIONS.

Study	Setup	Reported result	Relevance to AlphaHand
Mohamed et al. [9]	EEG wrist/finger movement imagery and execution; ICA, ERD/ERS features, MD/ANN classifiers.	65% MD and 71% ANN average accuracy.	Coarse wrist-vs.-finger context without individual-finger decoding.
Liao et al. [8]	128-channel EEG, 50-channel decoding set, ten usable subjects, pairwise one-hand finger decoding.	77.11% average pairwise EEG accuracy; resting-control data near guessing.	High-density offline finger-pair benchmark.
Xiao and Ding [10]	128-channel EEG; spectral PCA compared with classic mu/beta PSD features.	90–91% movement-vs.-rest detection; 39.7% five-finger PC accuracy versus 23.3% mu/beta PSD.	Shows spectral structures carry finger-specific information with far richer sensing than Muse 2.
Ding et al. [11]	128-channel BioSemi EEG; EEGNet decoder; real-time robotic feedback for two- and three-finger motor-imagery and movement-execution tasks.	80.56% two-finger and 60.61% three-finger online motor-imagery decoding accuracy across 21 experienced BCI users.	Establishes noninvasive finger-level robotic control with research-grade EEG; AlphaHand extends finger-level command evidence into consumer EEG with a four-channel Muse 2 command path.
AlphaHand (this work)	Four-channel Muse 2, subject-specific CNN+LSTM, action-conditioned finger decoding, gated replay.	89.79% action accuracy, 85.96% non-REST finger accuracy, 95.37% pseudo-live non-REST precision.	Low-density, subject-specific, deployment-gated command-path demonstration.

tion, subject-specific CNN+LSTM training, calibration, deterministic evaluation, and optional robotic-hand actuation.

- A deployment-consistent evaluation protocol that reports held-out accuracy alongside command-path precision, event hit rate, latency, and false REST actuation under pseudo-live replay.
- A channel-ablation analysis showing which Muse electrodes preserve subject-specific decoding performance, with temporal channels identified as the primary carriers of usable signal.
- A provenance account for the featured 2-M16 result, including session merging, REST pruning, window counts, saved run outputs, and the full traceable evidence chain.

The March 19, 2026 2-M16 bundle is the featured subject-specific deployment result throughout.

II. METHODS

A. Pipeline Overview

To ensure reported deployment metrics are extractable directly from the archive, the pipeline separates acquisition, training, evaluation, and actuation into seven independently logged stages. Acquisition records raw Muse samples with event markers; window extraction converts recordings into labeled fixed-length examples; training saves model checkpoints and calibrated test predictions; deterministic evaluation computes metrics and command-path summaries; calibration review generates reliability and uncertainty outputs; live inference optionally sends gated commands to the robotic hand; and post-live review converts prediction logs into auditable segments. Figure 1 summarizes the offline and online paths used in this paper.

B. Data Acquisition and Persistence

1) *Human Subjects and Ethics*: Recordings were collected under IRB approval permitting enrollment of participants ages 16 and older. All participants provided written consent or

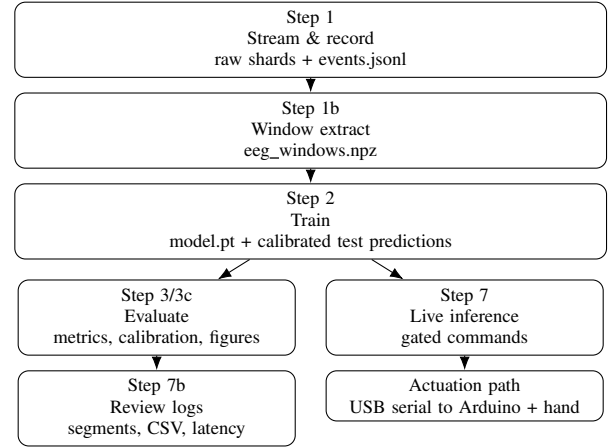


Fig. 1. End-to-end system flow and saved outputs. The online branch can transmit gated commands over USB serial to an Arduino-linked five-finger robotic hand. Quantitative results reported in this paper are extracted from archived stage outputs, with no manual transcription in the reporting chain.

assent; written parent or guardian consent was obtained for minors. The featured 2-M16 participant was a 16-year-old study-team self-subject. The broader project archive contains recordings from approximately 20 consented participants, ages 16–60, male and female, including students, teachers, and parents. This manuscript analyzes the featured 2-M16 deployment bundle because it contains the complete saved dataset, checkpoint, calibration, and replay evidence chain.

2) *Hardware*: The data analyzed here were acquired with a Muse 2 headset (Muse/InteraXon Inc., Toronto, ON, Canada), configured as a four-channel wearable EEG system at 256.0 Hz [17], [18]. The four electrodes (TP9, AF7, AF8, and TP10) place AF7 and AF8 over anterior-frontal sites and TP9 and TP10 over lateral temporal/periauricular sites within the international EEG placement system; Bird et al. provide a placement diagram relative to standard EEG landmarks [19]. Figure 2 also marks Fpz as the frontal reference point used to orient the headset montage. This hardware defines the central

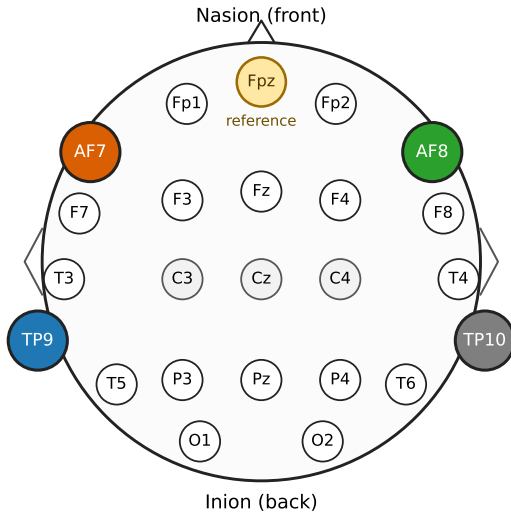


Fig. 2. Schematic of the Muse 2 electrode montage relative to simplified 10–20-style landmarks. AF7 and AF8 are anterior-frontal, TP9 and TP10 are lateral temporal/periauricular, and Fpz is marked as the frontal reference point. C3/Cz/C4 are included as motor-strip landmarks for orientation. The schematic is redrawn for this manuscript from standard placement descriptions and the Muse montage reference in Bird et al. [19].

constraint of the study: finger-level decoding from a consumer headset, operating without the central motor coverage of a research-grade cap [12], [13].

3) *EEG Artifact-Minimizing Recording Protocol*: During recording, subjects were seated, relaxed, and instructed to keep gaze fixed on the task point while avoiding non-task movements. The operator monitored stream health, electrode connectivity, and general system status live, with breaks used when needed. These controls were designed to reduce motion, gaze, and connectivity-related artifacts at recording time; artifact limitations are addressed in Section IV-A.

4) *Streaming Model*: All EEG samples are timestamped in the LSL clock domain [20] to prevent drift across devices and sessions. Event timestamps are converted into the same timebase before labeling. Absolute session time is defined as:

$$t_{\text{session}} = t_{\text{isl}} - t_{\text{stream_start}}$$

5) *Persistence Architecture*: Raw EEG samples are stored first as NumPy shards to preserve sample fidelity and avoid rounding error from CSV serialization. Event markers are written to both a legacy CSV view and an authoritative `events.jsonl` file. The analysis in this paper uses the JSONL event log as the canonical label source.

6) *Cohort Summary*: Demographics, dataset duration, and event label counts are extracted directly from saved metadata. Table II reports the featured participant metadata, and Table III reports dataset provenance. The featured 2-M16 run uses a filtered derived dataset assembled from two movement sessions and one auxiliary REST session; 522 of 12969 extracted windows were removed, leaving 12447 windows for

TABLE II
FEATURED PARTICIPANT DEMOGRAPHICS FROM REPOSITORY METADATA AND SUBJECT-CODE CONVENTION.

Subject	Age	Sex	Handedness
2-M16	16	Male	Right

the final dataset. Table IV reports the raw-session event counts underlying that dataset, and Fig. 3 summarizes the full data path for the featured 2-M16 result.

REST event IDs 0, 1, and 2 from session 2-M16_20260216_150056_01 were removed because they were repeatedly inferred as movement or accounted for the majority of held-out REST failures during dataset-quality review. This pruning criterion is recorded in the saved filter manifest. Because the featured training configuration points to the already-filtered derived dataset, the reported model was trained and evaluated only on data after this decision had been applied.

C. Window Extraction and Leakage Controls

Sliding windows are extracted with fixed duration 0.250 s and hop 0.050 s, then resampled to a fixed 256.0 Hz grid. Each window receives one action label and, when appropriate, one finger label. The deployable label space separates action from finger identity: a 5-way active-finger head is trained on non-REST windows, while a binary applicability head predicts whether a finger label is meaningful for the current window. Windows labeled with an active action (OPEN or CLOSE) but carrying finger NONE are rejected during extraction before downstream training or evaluation. Figure 4 grounds the procedure with an event-centered recorded segment and the corresponding overlapping extraction windows.

The nominal sample count per window is:

$$N = f_s \times T_{\text{window}} = 256.0 \times 0.250 = 64 \text{ samples}$$

Windows containing non-finite values are excluded during training. The extracted bundles also store extraction artifact flags and gap flags; the corresponding counts are summarized in Table V.

1) *Split Strategy and Leakage Controls*: A small number of REST events can still generate many overlapping REST windows under the sliding extraction scheme; REST window counts in the evaluation tables reflect this windowing effect. Training uses a group-aware 80/20 holdout split as recorded in each run configuration. The pipeline asserts no group overlap between train and test windows, supports optional temporal purging, and runs a window-index diagnostic to detect suspicious ordering leakage. In the representative saved outputs, the group-aware split is applied without an additional temporal exclusion band around the test windows.

The saved window bundles include trial and block identifier fields, but in the current outputs both are constant across each bundle. Strict trial-level evaluation is therefore unavailable for

TABLE III
REPRESENTATIVE DATASET PROVENANCE. THE FEATURED 2-M16 RUN IS TRAINED ON A FILTERED DERIVED DATASET ASSEMBLED FROM MULTIPLE RECORDED SESSIONS.

Subject	Dataset	Kind	Source	Notes
2-M16	Derived dataset (filtered)	Filtered derived	combined_20260317_211414	Pruned 522/12969 windows; Removed REST 0,1,2 from 20260216_150056

TABLE IV
RAW-SESSION EVENT LABEL COUNTS FOR THE RECORDED SESSIONS UNDERLYING THE FEATURED 2-M16 DATASET. ROLE INDICATES A MOVEMENT SESSION MERGED INTO THE DERIVED DATASET (CORE) OR AN AUXILIARY REST-ONLY SESSION ADDED FOR REST COVERAGE (AUX REST). COLUMN HEADERS ABBREVIATE THUMB/INDEX/MIDDLE/RING/PINKY OPEN AND CLOSE EVENTS.

Subject	Session	Role	REST	TH OPEN	TH CLOSE	IN OPEN	IN CLOSE	MID OPEN	MID CLOSE	RING OPEN	RING CLOSE	PINK OPEN	PINK CLOSE
2-M16	2-M16_20260216_150056_01	Core	5	58	55	54	57	55	59	57	60	60	59
2-M16	2-M16_20260317_190134	Core	10	1	2	2	3	3	2	1	3	2	3
2-M16	2-M16_20260315_145838_01	Aux REST	8	0	0	0	0	0	0	0	0	0	0

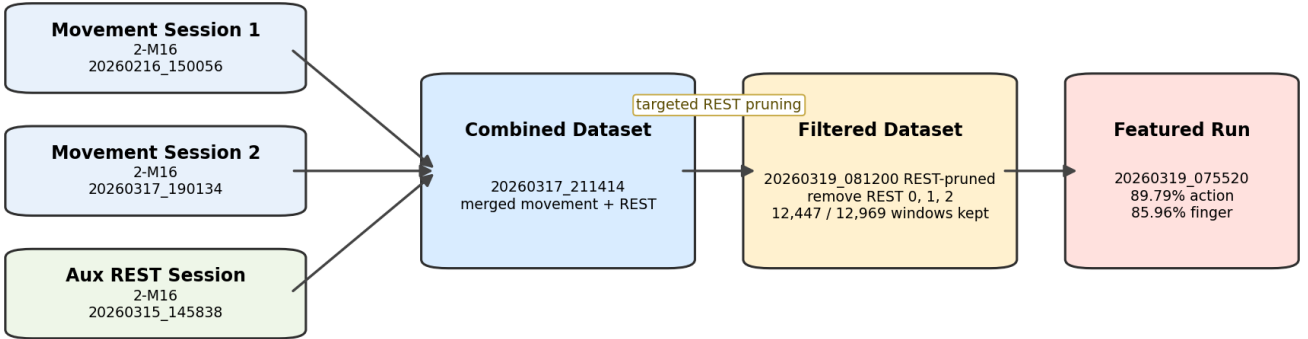


Fig. 3. Provenance of the featured 2-M16 result. Two movement sessions and one auxiliary REST session are merged into a combined dataset, targeted REST windows are removed according to the saved filter manifest, and the resulting filtered dataset produces the featured run used throughout the manuscript. The pruning step occurred before model training and evaluation.

TABLE V
DATASET AND WINDOW EXTRACTION SUMMARY FOR THE FEATURED 2-M16 DATASET REPORTED IN THIS MANUSCRIPT.

Subject	Session	N windows	Window (s)	Hop (s)	Mean overlap (%)	Extraction artifact flags	Gap flags
2-M16	Derived dataset (filtered)	12447	0.250	0.050	84.4	0	0

the runs reported here. The split logic instead relies primarily on event identifiers and secondary time-based grouping for windows with no event ID. Windows without event IDs arise primarily from REST/background intervals or boundary regions not assigned to a discrete movement event; these windows are grouped by secondary time-based identifiers so that temporally adjacent unlabeled-background windows remain together within a single split.

A post-hoc audit of the cached test indices found no shared (session_id, event_id) groups between test windows and the complementary training/calibration pool, and zero exact duplicate raw windows crossing the train/test boundary. This reduces the risk of direct split leakage; temporal depen-

dence from overlapping windows is a known property of the windowing scheme, addressed in Section IV-A.

2) *Diagnostics and Quality Controls*: The pipeline includes explicit hardening against common streaming-EEG failure modes: non-finite windows are removed before training, group-aware splitting prevents direct group overlap, optional purging can remove windows adjacent in time to test data, and a separate validation stage produces a dataset-quality report for each run.

D. Model and Training

AlphaHand uses a multi-head CNN+LSTM network with shared temporal features and three prediction heads: a 5-way active-finger head, a 3-way action head, and a binary

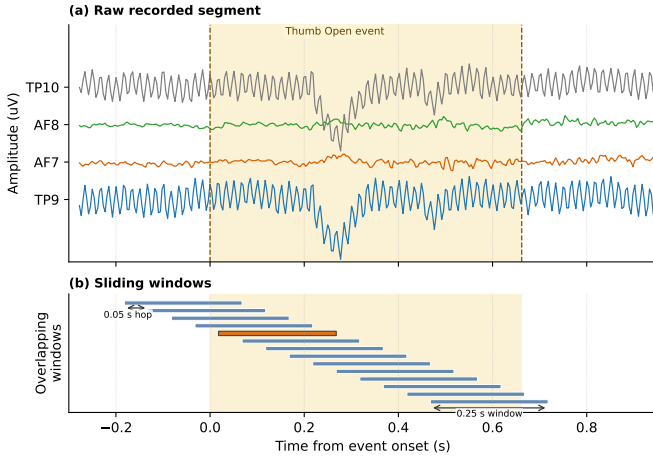


Fig. 4. Representative event-centered recorded segment from a source session underlying the featured 2-M16 dataset. (a) Raw Muse 2 traces around the labeled event. (b) Overlapping 0.250 s extraction windows with 0.050 s hop over the same interval. The overlap increases the number of labeled training examples, but it also induces temporal dependence between neighboring windows.

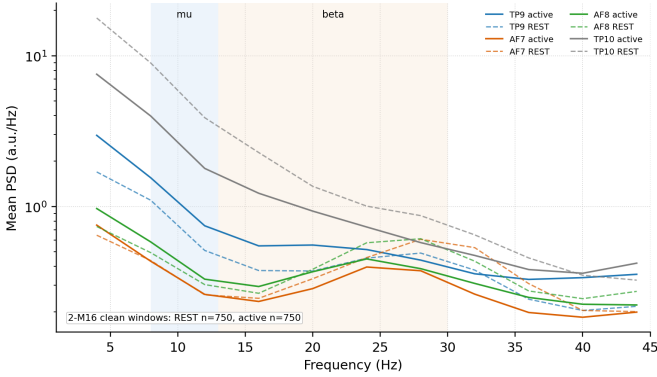


Fig. 5. Representative power spectral density from clean windows in the featured 2-M16 dataset. Solid traces average active-movement windows and dashed traces average REST windows for each Muse 2 channel. The panel is descriptive and is intended to ground the low-channel signal regime, without testing a preregistered spectral hypothesis.

finger-applicability head. The full architecture is summarized in Table VI. Two temporal 1D convolution blocks (16 and 32 channels; kernel sizes 5 and 3) feed into a single-layer LSTM with hidden size 64; each convolution block applies `GroupNorm(num_groups=min(4, Cout))` and dropout. The model has 27,673 trainable parameters. Before recurrent integration, the convolutional front end spans a 7-sample local temporal receptive field, approximately 27 ms at 256.0 Hz; the LSTM then integrates information across the full 0.250 s window.

The loss function is:

$$\mathcal{L} = \text{CE}_{\text{action}} + \lambda_f \text{CE}_{\text{finger}}(\text{non-REST}) + \lambda_a \text{BCE}_{\text{applicability}}$$

where the finger loss is computed only on non-REST windows and the applicability loss is computed on all win-

TABLE VI
CNN+LSTM MODEL CONFIGURATION FOR THE CURRENT DEPLOYABLE ACTIVE-FINGER SETUP.

Component	Setting
Input	$[T=64, C=4]$ EEG window
Conv1	16 channels, kernel 5, dropout 0.3
Conv2	32 channels, kernel 3, dropout 0.3
Norm	GroupNorm with $\min(4, C_{\text{out}})$ groups per convolution block
LSTM	1 layer, hidden size 64
Heads	Linear(64→5) active finger, Linear(64→3) action, Linear(64→1) applicability
Parameters	27,673

dows with target $1[\text{action} \neq \text{REST}]$. The action head additionally supports class weighting, including a configurable REST weight recorded in each run configuration. For the featured run, $\lambda_f = 1.0$ and $\lambda_a = 0.5$, as recorded in `train_config.json`; the action loss weight and REST action weight were both 1.0.

1) *Training Procedure*: Training hyperparameters are saved in `train_config.json`. The featured 2-M16 configuration uses 60 epochs, batch size 64, learning rate 0.001, and Adam optimization, with the archived configuration lacking a recorded learning-rate scheduler, early stopping, and gradient clipping. These are therefore not reported.

Input normalization is performed with a per-channel normalizer fit on the training split. For the featured run, the saved normalizer records per-window centering and detrending before channel normalization. A held-out calibration subset of the training data is then used to fit separate scalar temperatures for the action, finger, and applicability heads, and those parameters are saved alongside the trained model.

E. Calibration, Evaluation, and Actuation

1) *Monte Carlo Dropout for Uncertainty Estimation*: Deterministic held-out evaluation, deployment-consistent decoding, pseudo-live replay, and Step 7 live inference all run the network in evaluation mode with dropout disabled. Dropout is activated only for the separate Step 3c MC-dropout diagnostic used to visualize uncertainty. For that diagnostic, dropout layers are deliberately activated for K stochastic forward passes:

$$\mu = \frac{1}{K} \sum_{k=1}^K \hat{y}_k$$

$$\sigma^2 = \frac{1}{K} \sum_{k=1}^K (\hat{y}_k - \mu)^2$$

Reliability diagrams evaluate calibration, while confidence-uncertainty scatter plots visualize uncertainty concentration.

2) *Deterministic Evaluation and Calibration*: From each run's saved `test_predictions.npz`, Davanzo computed ECE with the same convention as the evaluation pipeline: 10 equal-width confidence bins over $[0, 0.1], (0.1, 0.2], \dots, (0.9, 1.0]$. In the archived predictions,

softmax confidences are strictly positive, so assigning the exact boundary value 0.0 to the first bin has no practical effect. Table VII reports action ECE over all test windows and finger ECE over true non-REST windows. The evaluation bundle also reports applicability false-positive and false-negative rates, action–applicability disagreement, and command-path invariants used to validate the deployment decoder.

For the featured run, action ECE is 0.0232 and finger ECE on true non-REST windows is 0.0273, indicating good agreement between predicted confidence and empirical accuracy.

3) *Online Inference and Actuation*: The command path is intentionally precision-biased: uncertain windows become no-send decisions before any best-guess actuation occurs. Postprocessing combines exponential smoothing, action/finger/applicability thresholds, a minimum joint-confidence requirement, a consecutive-window stability gate, and per-finger cooldown/hold intervals. In the featured March 19, 2026 bundle, commands require 3 agreeing windows, equal to 150 ms at a 0.050 s hop. Action, finger, and applicability thresholds are 0.05, 0.20, and 0.40.

The cooldown is an actuator-protection interval for the commanded finger, leaving other fingers free to receive commands if their own gates pass. The host serial ceiling is 20 commands/s, matching the 50 ms inference hop, and the Arduino receiver enforces per-finger hold timers with non-blocking servo updates. The action threshold is intentionally permissive; REST suppression is dominated by the REST/NONE invariant, finger and applicability gates, stability requirement, and cooldown.

When actuation is enabled, commands are sent over USB serial to an Arduino-linked five-finger robotic hand. Hardware safety gates prevent actuation whenever the predicted action is REST or the predicted finger is NONE, enforcing the same REST/NONE and OPEN-or-CLOSE/active-finger pairing used during deterministic evaluation.

Replay-time command-path summaries from the saved 2-M16 bundle form the auditable record for the results reported in Section III.

4) *Representative Run Selection and Reproducibility*: The project archive contains more recordings and runs than can be presented compactly. Quantitative reporting centers on the featured run because it includes archived windows, training configuration, deterministic evaluation, calibration review, and pseudo-live replay.

The March 19, 2026 2-M16 checkpoint was selected because the model-selection audit prioritized command-path safety—pseudo-live would-send precision, event hit rate, and false REST actuation—over offline split accuracy. A later checkpoint scored higher on some offline metrics, but its replay safety metrics were weaker.

The manuscript uses the archived window dataset, training configuration, summary metrics, calibrated predictions, replay output, and report figures for the featured 2-M16 run. The displayed training curve is a rerun of the featured recipe; the original deployment checkpoint predates per-epoch history logging.

III. RESULTS

This paper reports three levels of evaluation. First, held-out window metrics quantify action and finger separation under deterministic inference. Second, deployment-consistent decoding verifies that predictions remain valid after enforcing REST/NONE and active-finger invariants. Third, pseudo-live replay verifies that predictions survive the same stability gates, cooldowns, and command-suppression logic used before robotic actuation. Together, these levels establish the featured bundle’s command-path readiness under the archived deployment protocol.

Table VII reports the featured March 19, 2026 2-M16 deployment checkpoint. This run is the strongest complete evidence chain and anchors the analysis that follows. It reaches 89.79% action accuracy and 85.96% deployment-consistent finger accuracy on 2301 test windows (1994 true non-REST), well above uniform-prior chance levels of 33.3% for three-way action classification and 20% for five-way finger classification. The decisive result is that this performance survives the command logic required for actuation.

The paired deterministic evaluation and pseudo-live replay outputs inspect the full command path. Under deployment-style evaluation, the model reports 84.66% joint action–finger accuracy and 95.37% non-REST precision over 3324 emitted positive windows. Window-level send recall was 31.56%, a deployment-throughput metric reflecting gate behavior, with classifier error captured separately in the held-out window metrics. It denotes 3170 correctly emitted non-REST windows out of 10043 true non-REST replay windows and reflects overlapping windows, stability gates, command persistence, duplicate suppression, and cooldown. At the event level, replay hit 91.11% of 596 movement events, with 53 misses and 83 ms median first-command latency. Only 6 of 2404 true REST replay windows produced an actuation command, a false actuation rate of 0.25%.

A. Representative Validation Figures

For the featured 2-M16 run, the archived Step 3c reliability output includes the action and finger confusion matrices together with calibration summaries, while Table VII reports calibration error from the saved predictions. These views separate quantities that are often collapsed into a single headline number: class confusability, uncertainty concentration, and confidence calibration. Most action errors occur between OPEN and CLOSE, with fewer errors involving movement-vs.-REST separation, while finger errors concentrate among neighboring non-thumb digits, with THUMB remaining near ceiling. A supporting MC-dropout confidence–uncertainty diagnostic shows a dense region of high-confidence, low-uncertainty windows with a thinner diffuse tail, consistent with gate suppression of ambiguous commands before actuation. Figure 7 shows training dynamics for the featured recipe rerun, confirming stable convergence consistent with the archived deployment checkpoint.

TABLE VII
FEATURED 2-M16 DETERMINISTIC EVALUATION AND CALIBRATION SUMMARY. FINGER ACCURACY IS COMPUTED WITH THE DEPLOYMENT-CONSISTENT ACTION-CONDITIONED DECODE PATH ON TRUE NON-REST WINDOWS.

Subject	Run	Session	n_{test}	$n_{non-REST}$	Action Acc (%)	95% CI	Finger Acc _{non-REST} (%)	95% CI	Action ECE	Finger ECE _{non-REST}
2-M16	20260319_075520	Derived dataset (filtered)	2301	1994	89.79	[88.48, 90.96]	85.96	[84.36, 87.41]	0.0232	0.0273

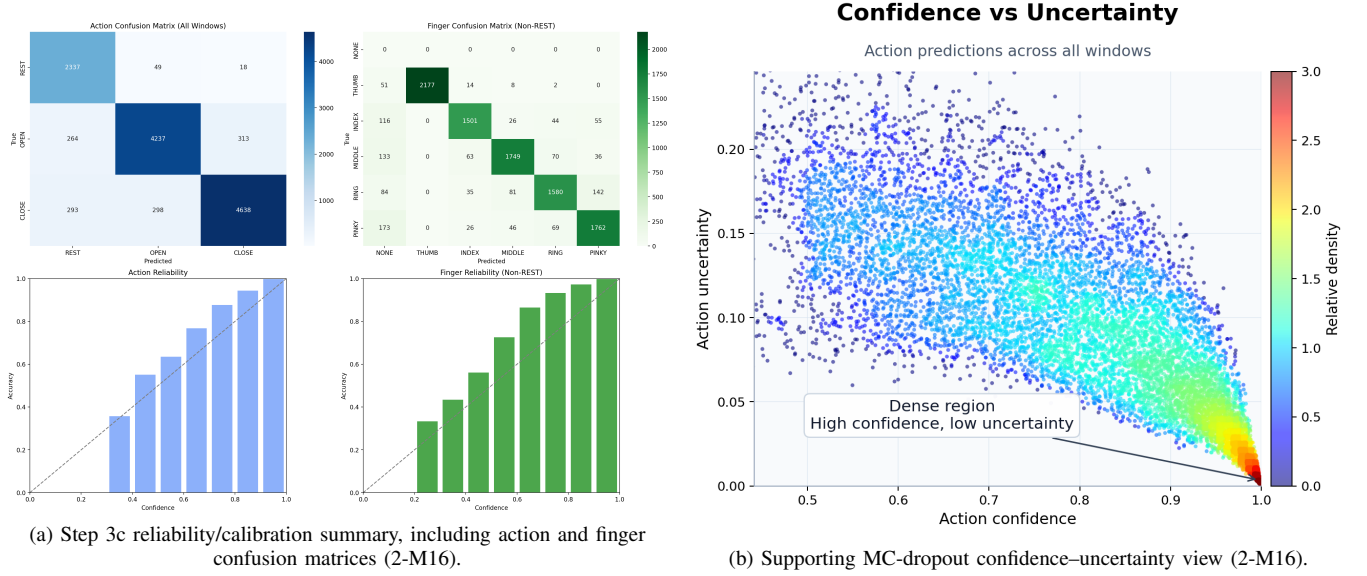


Fig. 6. Representative Step 3c evaluation outputs for the featured 2-M16 run. The reliability/calibration panel includes action and finger confusion matrices, while the confidence-uncertainty panel summarizes MC-dropout behavior across windows.

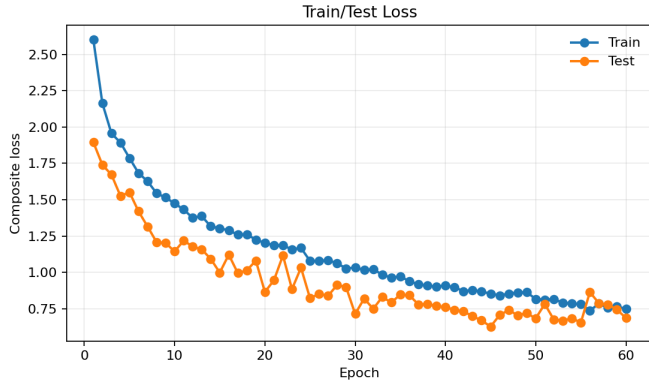


Fig. 7. Additional training-dynamics validation for the featured recipe: train/test composite loss by epoch (featured recipe rerun recorded history). The curve is generated only from recorded per-epoch history.

B. Per-Class Behavior

Per-class test accuracies and counts are derived from the saved per-window prediction bundles and reported separately for action and finger labels in Tables VIII and IX. The aggregate accuracy is distributed across classes, with no single class carrying the headline number. THUMB is near ceiling at 99.57%, INDEX remains strong at 94.50%, and PINKY reaches 89.57%, while MIDDLE and RING are lower at 79.01% and 65.48%, respectively. RING is the main outlier

at 65.48%, and its lower accuracy is best explained by finger-specific representational overlap, low-density channel placement, and subject-specific signal geometry, as discussed in Section IV. This asymmetry matters because a single average finger-accuracy value obscures which classes require further development.

Action-class results require the same disaggregated reading. For 2-M16, REST accuracy is high (98.37%) but is estimated from 307 REST test windows out of 2301 total test windows, so the denominator remains modest relative to the non-REST split. Continuous windowing expands the number of REST windows, but the REST subset remains small relative to OPEN and CLOSE. The pseudo-live false REST actuation rate is therefore reported separately from REST test accuracy because it directly measures false actuation under the replay command policy.

C. Exploratory Electrode Ablation

To identify electrode importance, Davanzo performed a post-hoc retraining ablation on the featured 2-M16 derived dataset. Each ablation run used the same subject-specific training recipe but changed which Muse channels were available during both training and testing. Importance was interpreted in two complementary ways: leave-one-out necessity, where a larger accuracy drop after removing one electrode indicates stronger dependence, and single-channel or pairwise sufficiency, where a reduced montage retaining high accuracy

TABLE VIII
ACTION-CLASS ACCURACIES FOR THE FEATURED 2-M16 RUN.

Subject	Run	Class	Count	Accuracy (%)
2-M16	20260319_075520	REST	307	98.37
2-M16	20260319_075520	OPEN	921	90.55
2-M16	20260319_075520	CLOSE	1073	86.67

TABLE IX
FINGER-CLASS ACCURACIES ON TRUE NON-REST TEST WINDOWS FOR THE FEATURED 2-M16 RUN, USING THE DEPLOYMENT-CONSISTENT ACTION-CONDITIONED FINGER DECODE PATH.

Subject	Run	Class	Count	Accuracy (%)
2-M16	20260319_075520	THUMB	466	99.57
2-M16	20260319_075520	INDEX	327	94.50
2-M16	20260319_075520	MIDDLE	443	79.01
2-M16	20260319_075520	RING	365	65.48
2-M16	20260319_075520	PINKY	393	89.57

TABLE X
EXPLORATORY 2-M16 CHANNEL ABLATION. FINGER ACCURACY IS NON-REST ACTIVE-FINGER ACCURACY FROM SAVED TEST PREDICTIONS, EVALUATED PRIOR TO DEPLOYMENT GATING. DROPS ARE PERCENTAGE-POINT CHANGES FROM THE FRESH FULL-MONTAGE RETRAIN.

Condition	Action (%)	Finger (%)
All channels	90.96	86.26
TP9+TP10 only	88.31	83.90
TP9 only	73.01	72.22
TP10 only	72.66	67.80
AF7 only	48.89	35.46
AF8 only	48.33	36.16
Drop TP9	-11.73 pp	-13.29 pp
Drop TP10	-12.08 pp	-9.13 pp
Drop AF7	-1.13 pp	+0.40 pp
Drop AF8	-0.56 pp	+1.30 pp

indicates that the retained electrodes carry usable decoding evidence.

The temporal electrodes dominate this ablation. Dropping TP9 or TP10 produces much larger accuracy losses than dropping AF7 or AF8, and each temporal electrode alone preserves substantially more non-REST finger accuracy than either frontal electrode alone. The TP9+TP10 temporal pair retains 88.31% action accuracy and 83.90% non-REST finger accuracy, only 2.65 pp and 2.36 pp below the full-montage retrain. In contrast, AF7-only and AF8-only finger accuracies fall to 35.46% and 36.16%, respectively. For this subject-specific split, the temporal/periauricular channels therefore provide substantially more usable EEG evidence for individual-finger decoding than the anterior-frontal channels.

D. Archive Scope

The project archive contains an earlier single-session 1-M16 run, but this manuscript excludes it from quantitative comparison because its dataset construction and evidence chain differ from the featured 2-M16 bundle. The next validation phase is prospectively paired multi-subject data collection

with matched dataset construction, filtering rules, and replay outputs across subjects.

E. Statistical Interpretation

This paper reports 95% confidence intervals (Wilson score) [21] for action accuracy over n_{test} windows and for finger accuracy over $n_{non-REST}$ windows in Table VII. Chance levels reflect uniform priors and serve as reference baselines in the presence of class imbalance. These intervals reflect finite test-set uncertainty under an i.i.d. window assumption while omitting temporal correlation induced by overlapping windows and event segments, which narrows the effective sample size. Percentile bootstrap intervals computed from the same saved correctness vectors closely tracked the Wilson intervals and did not change the interpretation, so the main paper reports a single interval family.

IV. DISCUSSION

AlphaHand preserves finger-level decoding through actuation-grade command logic. The featured run reaches high offline accuracy, preserves 84.66% deterministic joint action–finger accuracy after deployment-style decoding, and keeps false REST actuation to 0.25% under pseudo-live replay while hitting 91.11% of movement events. The deployment decoder reduces throughput, but the useful finger-level signal remains strong after the gates required for robotic-hand control.

The replay operating point favors precision over throughput. This is appropriate for an assistive-control command path because false movement can be more disruptive than missed movement. Precision is high (95.37%), while window-level send recall is lower than held-out accuracy because the denominator counts overlapping windows, leaving intended-movement counts to the event-level hit rate. In practical terms, the current gate limits false movements by suppressing uncertain commands; in the featured replay stream, only 6 actuation-sent windows occurred on 2404 true REST windows.

The practical usability target is event-level hit rate with first-command latency: did each intended movement receive at

least one correct command quickly enough to feel responsive? The current replay hit 91.11% of movement events with 83 ms median and 377 ms p95 first-command latency. Threshold relaxation, shorter stability requirements, and cooldown tuning provide direct throughput controls, with precision and false-actuation tradeoffs measured directly in replay and live testing.

The per-finger breakdown reveals a second important asymmetry. THUMB, INDEX, and PINKY are strong in the featured run, while MIDDLE and RING remain less stable. RING is the clearest example of this asymmetry. Its lower accuracy is best explained by overlap between middle/ring motor representations, mechanically coupled finger movement during data collection, and the absence of central motor-strip electrodes that would better separate neighboring digits. With only TP9, AF7, AF8, and TP10, none of the recording sites lies over the central hand area typically sampled by C3/C4/Cz-style motor electrodes [19]. The model is therefore decoding finger intent from a lateral frontal/temporal view of the sensorimotor system, operating without direct coverage of the central strip. The observed asymmetry reflects cortical representational overlap, limited channel coverage, and lower signal-to-noise ratio for some fingers, with classifier capacity as a secondary factor.

The channel ablation shows that the featured model relies primarily on temporal/periauricular information, with anterior-frontal channels contributing minimally to decoding performance. TP9 and TP10 are the most important electrodes by both leave-one-out and sufficiency criteria: removing either temporal electrode causes the largest action and finger losses, while the TP9+TP10 pair preserves most of the full-montage result. This shows where the trained classifier found the most useful subject-specific evidence in the available four-channel montage.

The comparison with Xiao and Ding’s 39.7% five-finger result is mechanistic, informing architecture without benchmarking performance. Their high-density study used a different task, feature family, and validation setting, while AlphaHand reports subject-specific, action-conditioned active-finger accuracy and then imposes REST/NONE command logic. The action/finger factorization, REST definition, and actuation gates change both the numerator and denominator. Xiao and Ding show that EEG can carry finger-specific spectral structure under rich sensing, while AlphaHand demonstrates that sufficient subject-specific structure remains in a sparse consumer montage to drive a precision-biased command path.

The next study phase will lock preprocessing, pruning, model-selection, calibration, and replay policies before data collection, then test command-path precision and event-level responsiveness across unfamiliar users and repeated sessions.

A. Limitations

Several limitations bound the current evidence.

Statistical limitations. Overlapping windows create temporal dependence, so Wilson intervals assume independent windows that the extraction procedure does not fully provide. Informative trial IDs are unavailable in the saved bundles,

preventing trial-level aggregation and cleaner effective-sample-size estimates. The effective sample size is therefore likely smaller than the raw window count.

Dataset limitations. Quantitative results are reported for one complete deployment evidence chain. The broader consented archive is more diverse, but those recordings do not yet provide matched datasets with trained checkpoints, calibration outputs, and replay outputs. The featured 2-M16 participant was a study-team self-subject, so familiarity with the task and recording protocol may inflate performance relative to users unfamiliar with the protocol. The featured dataset also uses targeted REST pruning, and the representative checkpoint was selected retrospectively from an archive using evidence-chain completeness and deployment safety. The channel ablations are single-seed, offline retraining studies; they provide subject-specific electrode-importance evidence, without informing anatomical localization. Prospective locked-protocol evaluation is needed before treating these numbers as unbiased multi-subject performance estimates.

Physiological and EEG artifact limitations. Muse 2 lacks dedicated EMG/EOG channels, and no post-hoc ICA-based artifact rejection was available. Protocol-level artifact reduction and stream-health monitoring reduce but cannot fully rule out non-cortical artifacts such as movement, muscle, gaze, or connectivity artifacts. The low-density electrode placement also does not directly cover C3/C4/Cz motor areas, limiting anatomical specificity for neighboring fingers.

Deployment limitations. Pseudo-live replay approximates live conditions, leaving full live-loop timing unvalidated. Dedicated live-loop latency logs are *not available*; replay-time summaries therefore serve as provisional timing evidence pending a full end-to-end audit. User-level command utility has not yet been tested prospectively.

V. CONCLUSION

AlphaHand demonstrates that consumer EEG can support more than coarse grasp detection in a subject-specific setting. The strongest run reaches 89.79% held-out action accuracy and 85.96% deployment-consistent finger accuracy, and the same bundle remains strong under deterministic evaluation and pseudo-live replay. The channel ablation further shows that, for this subject-specific split, TP9 and TP10 retained most of the full-montage decoding performance. This establishes an auditable end-to-end finger-level command-decoding path from a four-channel consumer headset. The next validation stage will test prospective multi-session performance with artifact-rejection logging, live-loop timing, false-actuation tracking, and user-level command-utility measures.

DATA AND CODE AVAILABILITY

Code, trained outputs, and derived datasets will be made available where consent and privacy constraints permit. Raw recordings containing participant EEG are governed by the study consent protocol and are not released without appropriate safeguards.

AUTHOR CONTRIBUTIONS AND FUNDING

Jonathan Davanzo served as first author and lead researcher for AlphaHand. He personally funded the project and led the project conception, software implementation, system construction, experimental pipeline, data analysis, AI/ML model development, model training and evaluation workflow, robotic-hand command path, hardware integration, and manuscript preparation. Liam Bales and Oliver Buchanan contributed to data collection by using the AlphaHand system developed by Davanzo during recording sessions. All authors used the system during the project. Bales and Buchanan reviewed the abstract, and full-paper review by all authors is planned before external submission.

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